A new Method of Image Encryption/Decryption via Voice Features

Zhiwei He, Yuanyuan Liu, Xueyi Ye, Haibin Yu
College of Communication Engineering
Hangzhou Dianzi University
Hangzhou, China

Abstract— **A new framework of image encryption via voice feature is proposed. Voice data is preprocessed and the MFCC coefficients are firstly calculated on each sub-frame; an iterated vector quantization is then applied on the MFCC data to obtain the key to image encryption and decryption. Image encryption and decryption is done in the frequency domain with a simple XOR operation. Due to the uniqueness and long length of the encryption key, along with the frequency domain encryption, the safeness of the whole system is assured.**

Keywords-image encryption; MFCC; voice feature

I. INTRODUCTION

Image encryption is widely used for secure transmission, privacy protection, etc. Its application fields include military, Internet, digital camera, etc. Traditional encryption algorithms, such as the DES algorithm whose key length is 56 bits, are not suitable for image encryption, because they have too short key lengths, which are weak for a brute-force attack. On the other hand, the RSA algorithm can provide a key with a length of 1024 bits; however its encryption algorithm is very complex which leads to a too slow speed to meet the fast encryption speed requirement for those data such as images.

Recently, many researchers successfully utilize the chaotic theory for image encryption[1][7-9]. For example, Yen and Guo [1] proposed a chaotic key-based algorithm (CKBA) for image encryption. In their system, the key, which is a binary sequence, is firstly generated using a chaotic system, the image pixels are then rearranged according to the generated binary sequence, the scale-gray values of pixels are then XORed or XNORed bit-by-bit to one of the two predetermined keys. In [2], Chen et al. proposed a symmetric image encryption schema based on chaotic system, in which two chaotic maps were used: a 3D chaotic map was exploited to shuffle the positions of image pixels, while another chaotic map was also used to confuse the relationship between the original image and its encrypted image. Other image encryption methods can be found in [10][11].

As stated above, the two key questions to image encryption is encryption key generation and encryption algorithm design. The encryption key must be unique and its length should not be too short. The designed encryption algorithm should not be too complex to be used for real-time applications. In this paper, we utilize the human voice feature as the encryption key to encrypt images. Being a kind of biological characteristic, the voice feature is very unique so that we can recognize the speaker or the spoken words through it. Use this unique feature as the encryption key can greatly assure the security of the whole image encryption system.

Many kinds of voice features extraction methods have been proposed in the literature. The Linear Prediction Cepstrum Coefficients (LPCC) and the Mel Frequency Cepstrum Coefficients (MFCC) [3] are two kinds of most widely used voice features for speaker recognition and voice recognition. As a kind of cepstrum feature, the LPCC is obtained by linear prediction with the assumption that the voice signal is auto regressive. The LPCC feature depicts one's unique voice track characteristics. The MFCC feature is the cepstrum coefficients extracted in the Mel scale frequency domain. Because it adequately models the characteristics of human being's sense of hearing without any assumption, the MFCC feature has better performance on recognition and noise resisting.

In order to obtain a stable model, the Cepstrum Coefficients can be modeled in many different ways, such as the Vector Quantization (VQ) method [4], the Hidden Markov Model (HMM) [6], the Artificial Neural Networks (ANN) [5], etc. The VQ method is a nonparametric, while the other two are parametric. A model obtained by VQ is much simpler with a shorter training time and a faster decision making, while the performance is quite the same to the other models.

In this paper, an iterated vector quantization algorithm is applied to the MFCC features of the input training voice data, the obtained data bits is used for the key to encrypt and decrypt images.

Fig.1 depicts the whole image encryption schema. Firstly, the input training voice data is preprocessed for denoising and silence elimination. The MFCC coefficients are then calculated on each sub-frame; an iterated vector quantization is then applied on the MFCC data to obtain the key to image encryption. Image encryption is done in the frequency domain with a simple XOR operation.

Fig.2 depicts the whole image decryption schema. Firstly, the user should offer his/her voice to make identity verification. If verified, the encrypted image is XORed bit by bit with the same encryption key to the encryption process; finally the encrypted image is decoded.

The work is partially supported by NSFC with Grant No. 60802047

978-1-4244-4131-0/09/$25.00 ©2009 IEEE

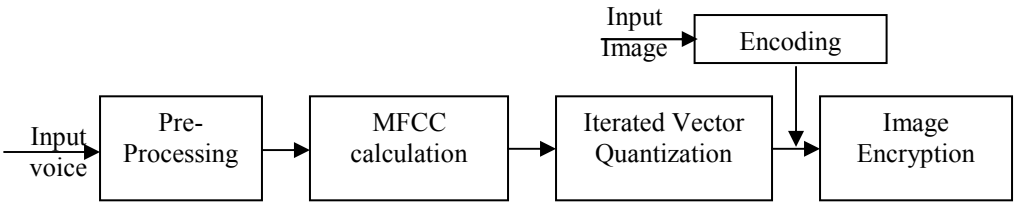


Figure 1. Image Encryption Schema based on voice features

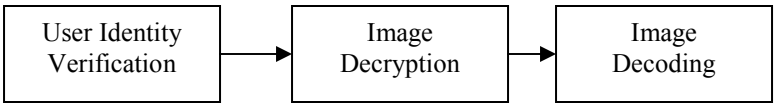


Figure 2. The decryption process of an encrypted image

The paper is organized as follows. Section I is an introduction. The calculation process of the MFCC data and its iterated vector quantization method is given in section II. Section III shows the image encryption and decryption process. Lastly, in section IV, results and conclusions will be given.

II. ITERATED VECTOR QUANTIZATION OF THE MFCC COEFFIENCIENTS

As mentioned before, in this paper, the voice feature is extracted as the encryption key. The main feature used is the MFCC data of the input voice. Before calculation of the MFCC data, denoising, silence detection and silence elimination should be firstly done on the input voice data in order to obtain stable features. The MFCC data is not static; that is to say, some differences exist for different input voice data even though the input data is from the same person. So the MFCC data should be quantized and clustered that we can use it as the encryption key. An iterated vector quantization method is applied with the MFCC data.

A. Preprocessing of input voice data

In order to decrease the noise level, a low-pass filter is firstly applied to the input voice data.

Because all silence voice has the same MFCC data, it should be eliminated first from the input voice data. In this paper, we simply use an energy threshold to separate the continuous silence data from other voice data. Generally, the input voice data can be separated to three parts, where the first part and the last part are silence data and only the second part contains the useful voice, as depicted in Fig. 3(a). So after silence detection and silence elimination, only the useful voice is used for the train data for MFCC calculation, as depicted in Fig. 3(b).

After that, the signal is filtered by the following digital filter:

$$y(n) = x(n) - \alpha x(n-1) \tag{1}$$

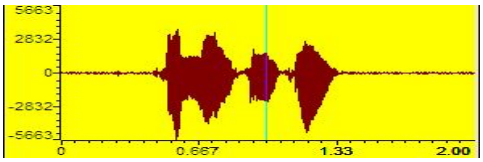


Fig.3 (a) input signal

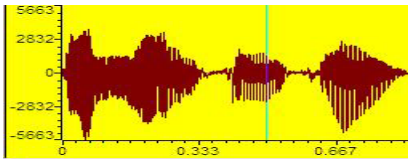


Fig.3 (b) extracted train signal

Figure 3. Silence detection and elimination

where $x(n)$ is the input sequence, $y(n)$ is the output of the filter, α is the pre-weight coefficient, in this paper, α is 0.97. The effect of this digital filter is to pre-weight the high frequencies of the input voice, which is more sensitive for voice recognition.

After being pre-weighted, the input sequence is cut and windowed to form sequential frames for feature extraction. Generally, an overlay exists between successive frames and the frame length should not be too large. In this paper, the frame length is chosen to be 256 and an overlay of half of the frame length is used. The raised cosine function is used as the window function, which has the form of the following:

$$y(n) = x(n) \times (0.54 - 0.46\cos(2\pi n(L-1))),$$

$$0 < n \le L-1 \tag{2}$$

where L is the frame length.

B. Calculation of the MFCC data

Each frame of voice data is used for calculation of the MFCC data.

The following steps can calculate the MFCC:

- For each frame of voice data $y(n)$, $0 \le n \le L-1$, a FFT transform is done on it and the obtained corresponding spectrum is $Y_a(k)$.
- The energy spectrum of $y(n)$ is filtered by a Mel-spaced triangular band-pass filter bank $\{H_m(k)\}$, and the logarithmical output energy for each of the filter is obtained according to the following equation:

$$S(m) = \ln\left[\sum_{k=0}^{L-1}|Y_a(k)|^2 H_m(k)\right],$$

$$0 \le m < M \tag{3}$$

where M is the number of filters in the filter bank, $H_m(k)$ is the frequency response of the $m-th$ filter.

- The Discrete Cosine Transform (DCT) is done with the sequence $\{S(m)\}$, and the obtained coefficients $\{C(n)\}$ are the MFCC data:

$$C(n) = \sum_{m=0}^{M-1} S(m)\cos[\frac{\pi n(m-0.5)}{M}],$$

$$0 \le n \le M-1 \tag{4}$$

C. Vector Quantization with Iteration

For each frame of voice data, a group of MFCC data $\vec{X}_i = \{C(n)\}$ can be obtained. As mentioned before, all groups of MFCC data should be processed further to obtain stable voice features. In this paper, one group of MFCC data is looked on as a vector and vector quantization is applied to each vector, the output of the quantization of a vector is a called a codebook, with each of its element named a code. The final stable codebook can be obtained by clustering with several iterations. The whole procedure is as follows:

- Because there are M MFCC coefficients, a nature way is to assume that a codebook also contains M codes. Select an initial codebook $B_M(0)$, do the following steps according to the input train data $\{\vec{X}_i\}$.
- At iteration i , find the best segmentation boundaries to segment the codebook according to the minimum average distortion rule. The updated codebook $B_M(i)$ is selected as the centroid of each segmented region.
- If the distortion is less than some preset threshold ε , the iteration ends and the obtained codebook is the final codebook, otherwise the iteration goes on until convergence or the maximum iteration reached.

The initial codebook $B_M(0)$ has great effect on the final codebook. A good method is to use the k-means algorithm to get it.

III. IMAGE ENCRYPTION AND DECRYPTION

In this paper, the whole set of codebooks $\{B_M(i), i = 1,2,...,Q\}$ obtained in the previous section are all used as the encryption key, where Q is the total iterations. As mentioned before, one codebook contains M codes. So the encryption key contains totally $Q \times M$ codes, where each code is a float with the size of 32bit. Apparently, with a proper selection of Q and M , the encryption key is long enough from being cracked with brute force.

A. Image Encryption

As shown in Fig.4, image encryption is done with two steps. The image is firstly encoded according to the JPEG standard. Then the encoded image is XORed with the encryption key bit by bit. When there are not enough bits in the encryption key, the encryption key is reused from the beginning. Because the JPEG encoding is based on the frequency domain and its file format is very special, it is very hard to crack the encrypted image although the encryption process is only a simple XOR operation.

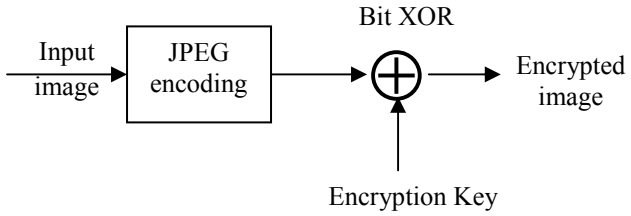


Figure 4. Flow chart of Image encryption

B. Image Decryption

The image decryption can be simply done with the following steps: firstly, the user should offer his/her voice to make identity verification, the flowchart of which is shown in Fig.5. As can be seen, the same voice feature is extracted and compared to the final codebook of the training voice. If the distance of them is small enough they match with each other. When matching occurs, the encrypted image I is XORed bit by bit with the stored encryption key Key obtained from the training voice.

From the above statements, we can have the following equations:

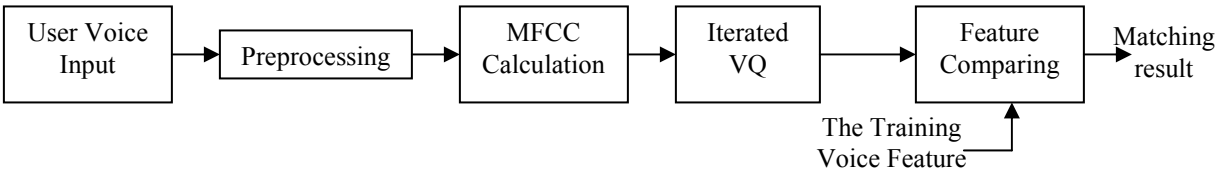


Figure 5. Flow chart for Identity verification

$$J = I \quad BitXOR \quad Key \tag{5}$$

we also have

$$I = I_o \quad BitXOR \quad Key \tag{6}$$

where I_o is the encoded input image for encryption. Substitute Eq. (6) to Eq. (5), we can get that

$$J = I_o \tag{7}$$

this means that the decrypted image is the same to the encoded version of the original input image.

The decrypted JPEG image is then decoded and the user can see it again.

IV. Results and Conclusions

When Q and M are selected to be 16 and 20 respectively, the extracted encryption key is 1024 bits long. The length of the encryption key can be made even longer with a larger Q and M . About 2 seconds of voice data with a sample rate of 8000Hz are used both for the training and verification. The success rate of voice feature matching is about 99%, which is high enough for practical applications. Images can be encrypted and decrypted successfully.

Similar to other biological features of human being, such as the face, the iris, etc., the voice feature is very unique. Almost every person has different voice features than others, so the extracted MFCC feature is very unique. That is, the encryption key for image encryption is unique. Secondly, a person's voice feature is relatively stable, that is, one can extract almost the same voice features at different time. This assures that the encrypted image can be almost always decrypted correctly, if the voice for encryption and decryption are from the same person. Finally, the extracted voice feature is so abundant that the encryption key can be very long, which make it very hard to decrypt the encrypted image, even with a brute-force manner.

The proposed method can be used on digital cameras and digital vidicons. In fact, the voice feature can also be used in other applications when encryption is needed. One problem with voice is that it can be recorded easily, which may lead to security problems in some occasions, e.g. the voice feature may not be suitable for bank card encryption and decryption. But for image encryption, there should be no problem at all.

References

[1] J. C. Yen and J. I. Guo, "A new chaotic key-based design for image encryption and decryption," Proc. IEEE Int. Conf. ircuits and Systems, vol. 4, pp. 49-52, 2000.

[2] G. R. Chen, Y. Mao and C. K. Chui, "A symmetric image encryption based on 3D chaotic maps," Chaos, Solitons & Fractals, vol. 21, no. 3, pp. 749-761, 2004.

[3] H.Torres, H.Rufiner, Automatic. "Speaker Identifacation by Means of Mel Cepstrum, Wavelets and Wavelet Packets." Processing of the 22th Annual EMBS International Conference, Chicago. July 2000. Page 978-981.

[4] Ningping.Fan, Justinian.Rosca. "Enhance VQ-Based Algorithms for Speech Independent Speaker Identification." AVBPA. 2003. Page 470-477.

[5] Siew Chan Woo, Chee Peng Lim, Osman.R. "Developmemt of a Speaker Recognition System Using Wavelets and Artificial Neural Network." Intelligent Multimedia, Video and Speech Processing May 2001.. Pages: 413-416.

[6] Qiang Huo, Chin-Hui Lee. "On-line adaptive learning of the correlated continuous density hidden Markov models for speech recognition." IEEE Transactions on Speech and Audio Processing, July 1998. Page(s):386 - 397.

[7] De Wang, Yuan-Biao Zhang. "Image Encryption Algorithm Based on S-boxes Substitution and Chaos Random Sequence**."** ICCMS '09. International Conference on Computer Modeling and Simulation, 2009. Pages: 110 – 113.

[8] Li Chuanmu, Hong Lianxi. "A New Image Encryption Scheme based on Hyperchaotic Sequences**."** IEEE International Workshop on Anti-counterfeiting, Security, Identification, 2007 .April 2007 Page(s):237 – 240.

[9] Shujun Li, Xuan Zheng. "On the security of an image encryption method**".** 2002 International Conference on Image Processing. Sept. 2002 Page(s):II-925 - II-928 vol.2 .

[10] Yi Xin, Ran Tao, Yue Wang. "Image Encryption Based on a Novel Reality-Preserving Fractional Fourier Transform.**"** First International Conference on Innovative Computing, Information and Control, 2006. ICICIC '06. Aug. 30 2006-Sept. 1 2006 Page(s):22 – 25.

[11] Shiguo Lian; Jinsheng Sun; Zhiquan Wang;"A novel image encryption scheme based-on JPEG encoding.**"** Eighth International Conference on Information Visualisation, 2004. IV 2004. July 2004 Page(s):217 – 220.